

7(a)(i)	The engine takes in air at a rate of 1300 kg s^{-1} $\frac{\Delta m}{\Delta t} = \rho A \frac{\Delta s}{\Delta t}$ $= \rho \left(\frac{\pi D^2}{4} \right) v_{\text{in}}$ $v_{\text{in}} = \frac{1300}{1.2 \times \frac{\pi}{4} \times 3.0^2}$ $= 153.3$ $\approx 150 \text{ m s}^{-1} \text{ (shown)}$	[1] for correct substitution [1] for correct calculated value before 150 m s^{-1}
7(a)(ii)	Thrust, $F = (m_{\text{air}} + m_{\text{fuel}}) v_{\text{exhaust}} - m_{\text{air}} v_{\text{in}}$ $= (1300 + 17)(410) - (1300)(150)$ $= 344970$ $\approx 345 \text{ kN}$	[1] for correct substitution [1] for correct answer
7(a)(iii)	By Newton's second law, $F = ma$ $a = \frac{F}{m}$ $= \frac{344970 \times 2}{2.75 \times 10^6} \text{ (because of 2 engines)}$ $= 2.461$ $\approx 2.46 \text{ m s}^{-2}$	[1] for correct substitution [1] for correct answer
7(b)	As air is deflected upwards upon contact with the stabiliser, it experiences a rate of change of momentum. By Newton's second law, this is the resultant force on the air by the stabiliser.	[1] for application of Newton's 2nd law

No.	Solution	Remarks
	By Newton's third law, the air will exert a force of equal magnitude but in opposite direction, causing a downward force on the stabiliser and enabling the aircraft to take off.	[1] for application of Newton's 3rd law
7(c)(i)	Consider vertical forces in equilibrium, $N_R + N_F = W$ $N_F = W - N_R$ Take moments about the centre of gravity, $N_R X_R = N_F X_F + Fd$ $= (W - N_R) X_F + Fd$ $N_R (X_R + X_F) = W X_F + Fd$ $\therefore N_R = \frac{W X_F + Fd}{X_R + X_F}$ (shown)	[1] for vertical forces in equilibrium [1] for correct expression taking moments [1] for substitution
7(c)(ii)	$N_F = W - N_R$ $= W - \frac{W X_F + Fd}{X_R + X_F}$ $= \frac{W(X_R + X_F) - W X_F - Fd}{X_R + X_F}$ $= \frac{W X_R - Fd}{X_R + X_F}$	[1] for correct substitution
7(c)(iii)	When at rest without the thrust F, the contact force on the front landing gears is $\frac{W X_R}{X_R + X_F}$ and the contact force on the rear landing gears is $\frac{W X_F}{X_R + X_F}$. When accelerating, the thrust F results in the contact force on the front gears to decrease by $\frac{Fd}{X_R + X_F}$, while the contact force on the rear gears is increased by $\frac{Fd}{X_R + X_F}$, causing the weight transfer from the front gears to the rear gears.	[1] for explanation of contact forces without acceleration [1] for explanation of the transfer of force from front gears to the rear gears
7(d)(i)	Total energy used	[1] for correct substitution

No.	Solution	Remarks
	$= (5.5 \times 10^3)(12)(42 \times 10^6)$ $= 2.772 \times 10^{12}$ $\approx 2.77 \times 10^{12} \text{ J}$	[1] for correct answer
7(d)(ii)	The amount of energy that the battery stores per kilogram of its mass is 0.260 kilowatt-hours.	[1]
7(d)(iii)	Mass of lithium battery $= \frac{2.772 \times 10^{12}}{0.260 \times 10^3 \times 60 \times 60}$ $= 2.962 \times 10^6$ $\approx 2.96 \times 10^6 \text{ kg}$	[1] for correct substitution [1] for correct answer
7(e)(i)	The total weight of the battery ($2.9 \times 10^7 \text{ N}$) will be greater than the maximum take-off weight (MTOW) of $2.75 \times 10^6 \text{ N}$ of the aircraft, which is not possible.	[1]
7(e)(ii)	Any two: 1. Jet fuel has a higher energy density (42 MJ kg^{-1}) compared to battery ($0.260 \text{ kW h kg}^{-1}$, i.e. $9.36 \times 10^5 \text{ J kg}^{-1}$), hence more suitable for long-haul flights. 2. Electric aircrafts need batteries with high enough specific energy, so the mass penalty means they cannot fly long distances. 3. Lithium batteries can overheat and ignite, posing fire hazard on electric aircrafts. 4. Jet engine aircraft uses jet fuel which gets depleted over the course of the flight causing the aircraft to be lighter and hence burns less fuel whereas the mass of the battery-powered electric aircraft remains constant throughout the flight.	[2] Any other logical answer